Solution: Problem Challenge 1: Permutation in a String

Problem Statement

Given a string and a pattern, find out if the **string contains any permutation of the pattern**.

Permutation is defined as the re-arranging of the characters of the string. For example, “abc” has the following six permutations:

1. abc
2. acb
3. bac
4. bca
5. cab
6. cba

If a string has ‘n’ distinct characters, it will have $n!$ permutations.

Example 1:

```
Input: String="oidbcaf", Pattern="ABC"
Output: true
Explanation: The string contains "bca" which is a permutation of the giv
en pattern.
```

Example 2:

```
Input: String="odicf", Pattern="dc"
Output: false
Explanation: No permutation of the pattern is present in the given strin
g as a substring.
```

Example 3:

```
Input: String="bcdxabcdy", Pattern="bcdyabcdx"
Output: true
Explanation: Both the string and the pattern are a permutation of each o
ther.
```

Example 4:

```
Input: String="aaacb", Pattern="abc"
Output: true
Explanation: The string contains "acb" which is a permutation of the giv
en pattern.
```

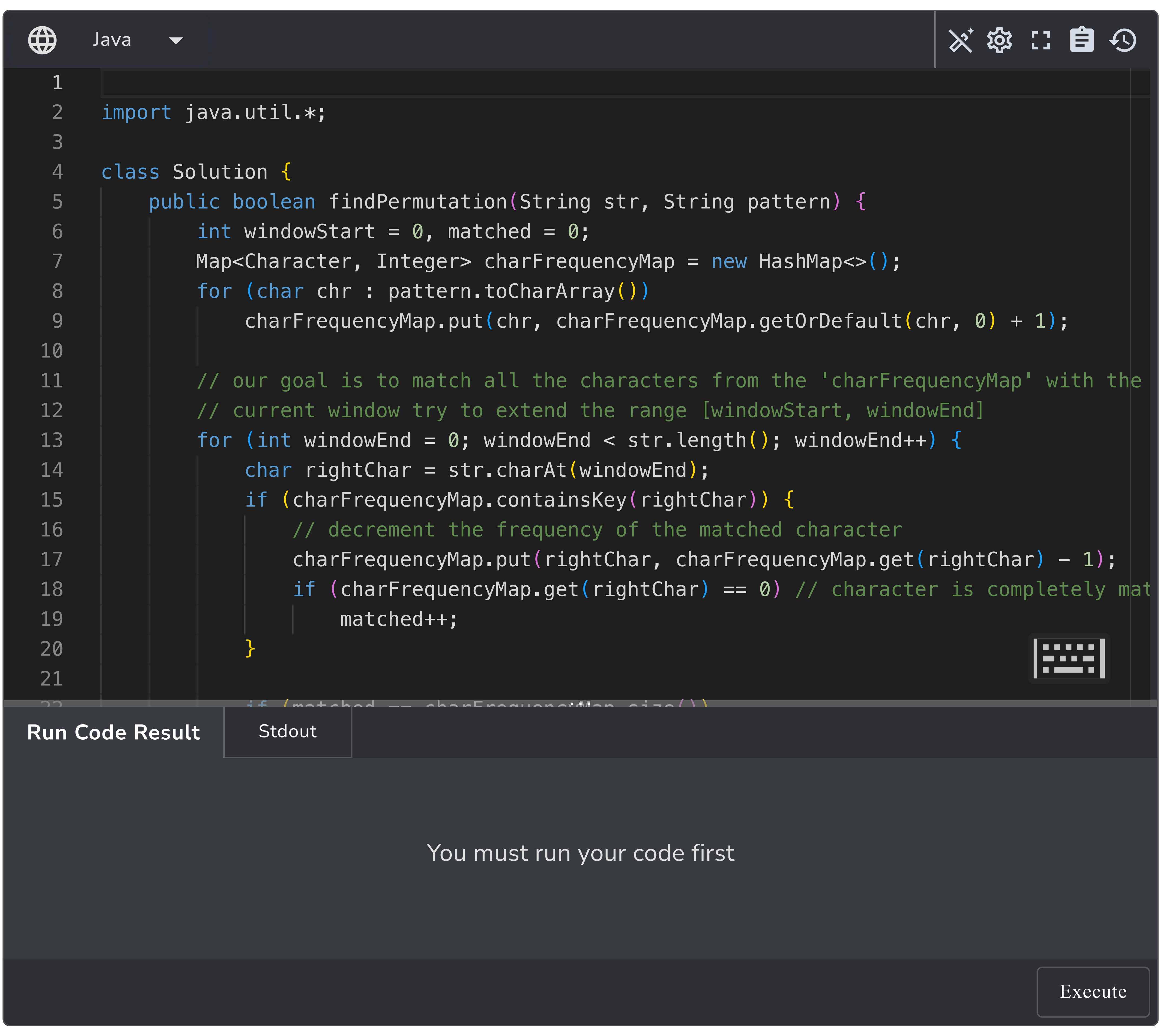
Solution

This problem follows the **Sliding Window** pattern, and we can use a similar sliding window strategy as discussed in Longest Substring with K Distinct Characters. We can use a **HashMap** to remember the frequencies of all characters in the given pattern. Our goal will be to match all the characters from this **HashMap** with a sliding window in the given string. Here are the steps of our algorithm:

1. Create a **HashMap** to calculate the frequencies of all characters in the pattern.
2. Iterate through the string, adding one character at a time in the sliding window.
3. If the character being added matches a character in the **HashMap**, decrement its frequency in the map. If the character frequency becomes zero, we got a complete match.
4. If at any time, the number of characters matched is equal to the number of distinct characters in the pattern (i.e., total characters in the **HashMap**), we have gotten our required permutation.
5. If the window size is greater than the length of the pattern, shrink the window to make it equal to the pattern’s size. At the same time, if the character going out was part of the pattern, put it back in the frequency **HashMap**.

Code

Here is what our algorithm will look like:



```java

import java.util.*;

class Solution {
    public boolean findPermutation(String str, String pattern) {
        int windowStart = 0, matched = 0;
        Map<Character, Integer> charFrequencyMap = new HashMap<>();
        for (char chr : pattern.toCharArray())
            charFrequencyMap.put(chr, charFrequencyMap.getOrDefault(chr, 0) + 1);

        // our goal is to match all the characters from the 'charFrequencyMap' with the
        // current window try to extend the range [windowStart, windowEnd]
        for (int windowEnd = 0; windowEnd < str.length(); windowEnd++) {
            char rightChar = str.charAt(windowEnd);
            if (charFrequencyMap.containsKey(rightChar)) {
                // decrement the frequency of the matched character
                charFrequencyMap.put(rightChar, charFrequencyMap.get(rightChar) - 1);
                if (charFrequencyMap.get(rightChar) == 0) // character is completely mat
                    matched++;
            }

```

You must run your code first

Time Complexity

The above algorithm’s time complexity will be $O(N+M)$, where ‘N’ and ‘M’ are the number of characters in the input string and the pattern, respectively.

Space Complexity

The algorithm’s space complexity is $O(M)$ since, in the worst case, the whole pattern can have distinct characters that will go into the **HashMap**.

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